Analyzing voting trends in Tamil Nadu Assembly Elections (2011-2021)

1.Project Over view and Objective

This project involves cleaning, transforming, and analysing raw data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful business insights.

The main objective is to demonstrate data exploration and data cleaning techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

To analyze the voting trends and electoral changes in Tamil Nadu Assembly Elections from 2011 to 2021.

2.Data source:

Link : https://indiadataportal.com/search?page=1&sortBy=lastModified&sectors=Elections

India data portal offers downloadable datasets related to elections, including constituency-level results and voting statistics.

- **Domain: Election**

3.Problem Statement

- Analyze voter turnout (%) by year to identify trends
- Study youth voter participation and influence
- Examine gender-based winning trends (male vs female)
- Evaluate vote share percentage by party
- Analyze changes in party vote share over time
- Measure winning margins across constituencies
- Identify constituencies with highest vote share
- Count number of winners per party by year
- Analyze total valid votes across elections
- Compare constituency-wise party performance
- Study changes in gender representation over time

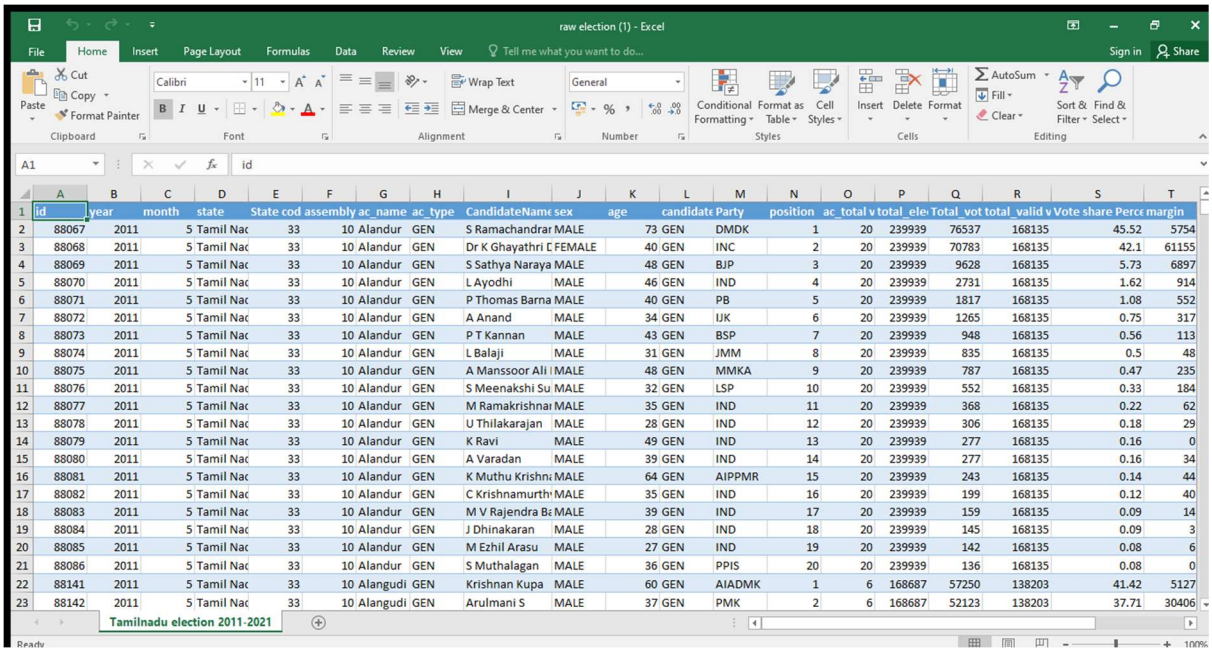
4.Column Name And Description:

Column Name	Data Type	Description
Election Id	Whole Number	Unique Id of election
Year	Whole Number W	Election conducted Year
Assembly Constituency name	Text	Each Constituency in Election
Assembly Constituency Type	Text	GEN, SC, ST
Candidate Name	Text	Name of the Candidate
Sex	Text	Sex of the Candidate
Age	Number	Age of the Candidate
Party	Text	Candidate Belongs to which Party
Vote Share Percentage	Decimal Number	Percentage of total vaild vote received by Candidate
Margin	Whole Number	Difference between winner and runner vote
Turnout Percentage	Decimal Number	Percentage of Turnout

5.Data Cleaning and Transformation Process using Excel:

Data Import

- Imported raw Excel sheets (2011–2021 election data)
- verified column headers and standardized naming conventions.



id	year	month	state	State cod	assembly	ac_name	ac_type	CandidateName	sex	age	candidate	Party	position	ac_total v	total_ele	Total_vot	total_valid v	Vote share	Perce	margin
88067	2011	5	Tamil Nac	33	10	Alandur	GEN	S Ramachandrar	MALE	73	GEN	DMDK	1	20	239939	76537	168135	45.52		5754
88068	2011	5	Tamil Nac	33	10	Alandur	GEN	Dr K Ghayathri D	FEMALE	40	GEN	INC	2	20	239939	70783	168135	42.1		61155
88069	2011	5	Tamil Nac	33	10	Alandur	GEN	S Sathya Naraya	MALE	48	GEN	BJP	3	20	239939	9628	168135	5.73		6897
88070	2011	5	Tamil Nac	33	10	Alandur	GEN	L Ayodhi	MALE	46	GEN	IND	4	20	239939	2731	168135	1.62		914
88071	2011	5	Tamil Nac	33	10	Alandur	GEN	P Thomas Barna	MALE	40	GEN	PB	5	20	239939	1817	168135	1.08		552
88072	2011	5	Tamil Nac	33	10	Alandur	GEN	A Anand	MALE	34	GEN	IJK	6	20	239939	1265	168135	0.75		317
88073	2011	5	Tamil Nac	33	10	Alandur	GEN	P T Kannan	MALE	43	GEN	BSP	7	20	239939	948	168135	0.56		113
88074	2011	5	Tamil Nac	33	10	Alandur	GEN	L Balaji	MALE	31	GEN	JMM	8	20	239939	835	168135	0.5		48
88075	2011	5	Tamil Nac	33	10	Alandur	GEN	A Manssoor Ali	MALE	48	GEN	MMKA	9	20	239939	787	168135	0.47		235
88076	2011	5	Tamil Nac	33	10	Alandur	GEN	S Meenakshi Su	MALE	32	GEN	LSP	10	20	239939	552	168135	0.33		184
88077	2011	5	Tamil Nac	33	10	Alandur	GEN	M Ramakrishnar	MALE	35	GEN	IND	11	20	239939	368	168135	0.22		62
88078	2011	5	Tamil Nac	33	10	Alandur	GEN	U Thilakarajan	MALE	28	GEN	IND	12	20	239939	306	168135	0.18		29
88079	2011	5	Tamil Nac	33	10	Alandur	GEN	K Ravi	MALE	49	GEN	IND	13	20	239939	277	168135	0.16		0
88080	2011	5	Tamil Nac	33	10	Alandur	GEN	A Varadan	MALE	39	GEN	IND	14	20	239939	277	168135	0.16		34
88081	2011	5	Tamil Nac	33	10	Alandur	GEN	K Muthu Krishna	MALE	64	GEN	AIPPMR	15	20	239939	243	168135	0.14		44
88082	2011	5	Tamil Nac	33	10	Alandur	GEN	C Krishnamurth	MALE	35	GEN	IND	16	20	239939	199	168135	0.12		40
88083	2011	5	Tamil Nac	33	10	Alandur	GEN	M V Rajendra Ba	MALE	39	GEN	IND	17	20	239939	159	168135	0.09		14
88084	2011	5	Tamil Nac	33	10	Alandur	GEN	J Dhinakaran	MALE	28	GEN	IND	18	20	239939	145	168135	0.09		3
88085	2011	5	Tamil Nac	33	10	Alandur	GEN	M Ezhil Arasu	MALE	27	GEN	IND	19	20	239939	142	168135	0.08		6
88086	2011	5	Tamil Nac	33	10	Alandur	GEN	S Muthalagan	MALE	36	GEN	PPIS	20	20	239939	136	168135	0.08		0
88141	2011	5	Tamil Nac	33	10	Alangudi	GEN	Krishnan Kupa	MALE	60	GEN	AIADMK	1	6	168687	57250	138203	41.42		5127
88142	2011	5	Tamil Nac	33	10	Alangudi	GEN	Arulmani S	MALE	37	GEN	PMK	2	6	168687	52123	138203	37.71		30406

Duplication:

- Removed duplicate rows based on Constituency Name + Year + Candidate Name.
- **ID Assignment**
- Created surrogate keys for Assembly ID, Candidate ID, Party ID.

Data Modelling

- Converted raw data into structured tables:

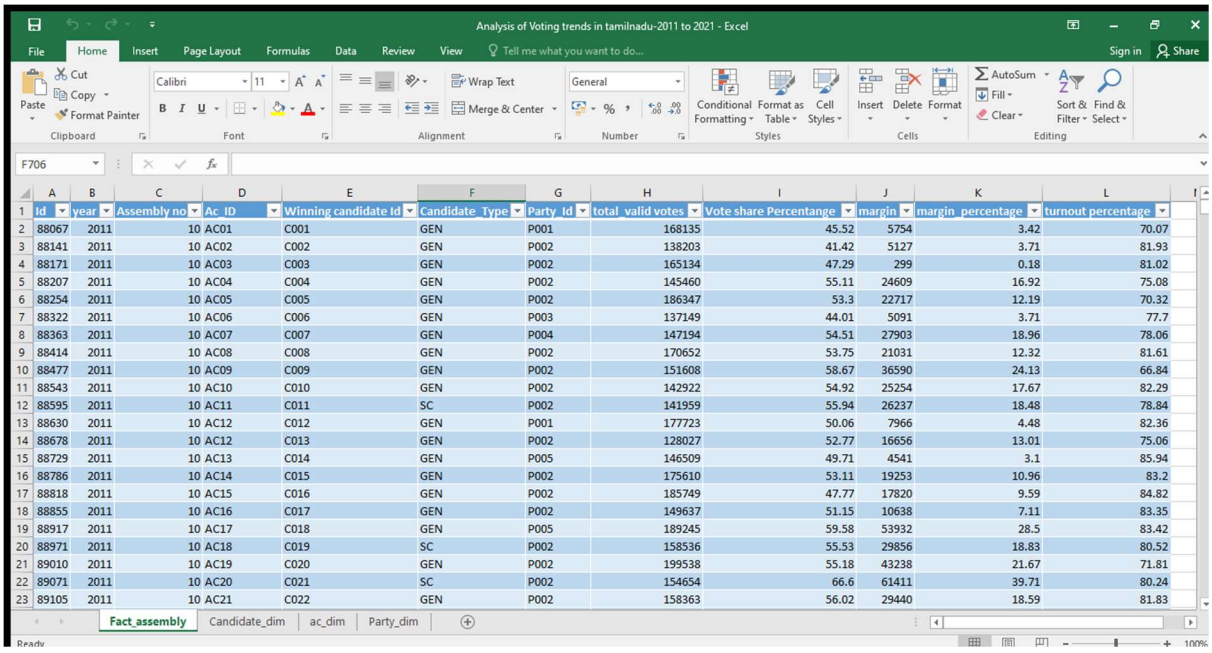
Fact Table Contain:

- Election Id
- Year
- Assembly No
- AC Id
- Candidate Id
- Party Id

- Total Valid Votes
- Vote Share %
- Margin
- Turnout %

Dimensional Table:

- **AC_Dim**: Constituency details
- **Party_Dim**: Party details
- **Candidate_Dim**: Candidate info
- **Year_Dim**: Year & term



Id	year	Assembly no	Ac_ID	Winning candidate Id	Candidate_Type	Party_Id	total_valid votes	Vote share Percentange	margin	margin_percentage	turnout percentage
88067	2011	10	AC01	C001	GEN	P001	168135	45.52	5754	3.42	70.07
88141	2011	10	AC02	C002	GEN	P002	138203	41.42	5127	3.71	81.93
88171	2011	10	AC03	C003	GEN	P002	165134	47.29	299	0.18	81.02
88207	2011	10	AC04	C004	GEN	P002	145460	55.11	24609	16.92	75.08
88254	2011	10	AC05	C005	GEN	P002	186347	53.3	22717	12.19	70.32
88322	2011	10	AC06	C006	GEN	P003	137149	44.01	5091	3.71	77.7
88363	2011	10	AC07	C007	GEN	P004	147194	54.51	27903	18.96	78.06
88414	2011	10	AC08	C008	GEN	P002	170652	53.75	21031	12.32	81.61
88477	2011	10	AC09	C009	GEN	P002	151608	58.67	36590	24.13	66.84
88543	2011	10	AC10	C010	GEN	P002	142922	54.92	25254	17.67	82.29
88595	2011	10	AC11	C011	SC	P002	141959	55.94	26237	18.48	78.84
88630	2011	10	AC12	C012	GEN	P001	177723	50.06	7966	4.48	82.36
88678	2011	10	AC12	C013	GEN	P002	128027	52.77	16656	13.01	75.06
88729	2011	10	AC13	C014	GEN	P005	146509	49.71	4541	3.1	85.94
88786	2011	10	AC14	C015	GEN	P002	175610	53.11	19253	10.96	83.2
88818	2011	10	AC15	C016	GEN	P002	185749	47.77	17820	9.59	84.82
88855	2011	10	AC16	C017	GEN	P002	149637	51.15	10638	7.11	83.35
88917	2011	10	AC17	C018	GEN	P005	189245	59.58	53932	28.5	83.42
88971	2011	10	AC18	C019	SC	P002	158536	55.53	29856	18.83	80.52
89010	2011	10	AC19	C020	GEN	P002	199538	55.18	43238	21.67	71.81
89071	2011	10	AC20	C021	SC	P002	154654	66.6	61411	39.71	80.24
89105	2011	10	AC21	C022	GEN	P002	158363	56.02	29440	18.59	81.83

Data Standardization

- Removed extra spaces
- Corrected spelling errors
- Standardized party names (e.g., DMK, AIADMK)
- ⬙ Handling Missing Values
- Filled missing values using averages/interpolation
- Maked unknown values as NULL

Transformation

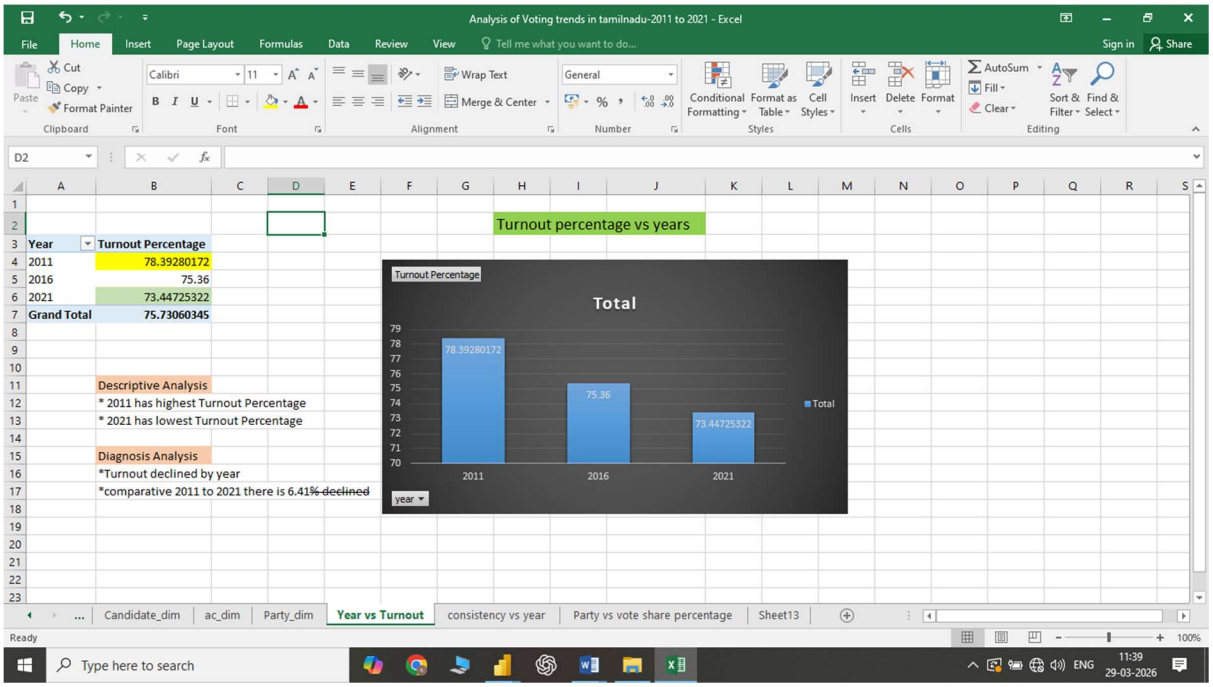
- Created calculated fields:
  - Vote Share %
  - Margin %

Validation **Formulas**

- **Vote Share %** = (Votes Received / Total Valid Votes) × 100
- **Margin** = Winner Votes − Runner Votes
- **Margin %** = (Margin / Total Valid Votes) × 100
- **Turnout %** = (Votes Polled / Total Electors) × 100

6.Pivot Chart Anlaysis in Excel

1.Voter Turnout Analysis



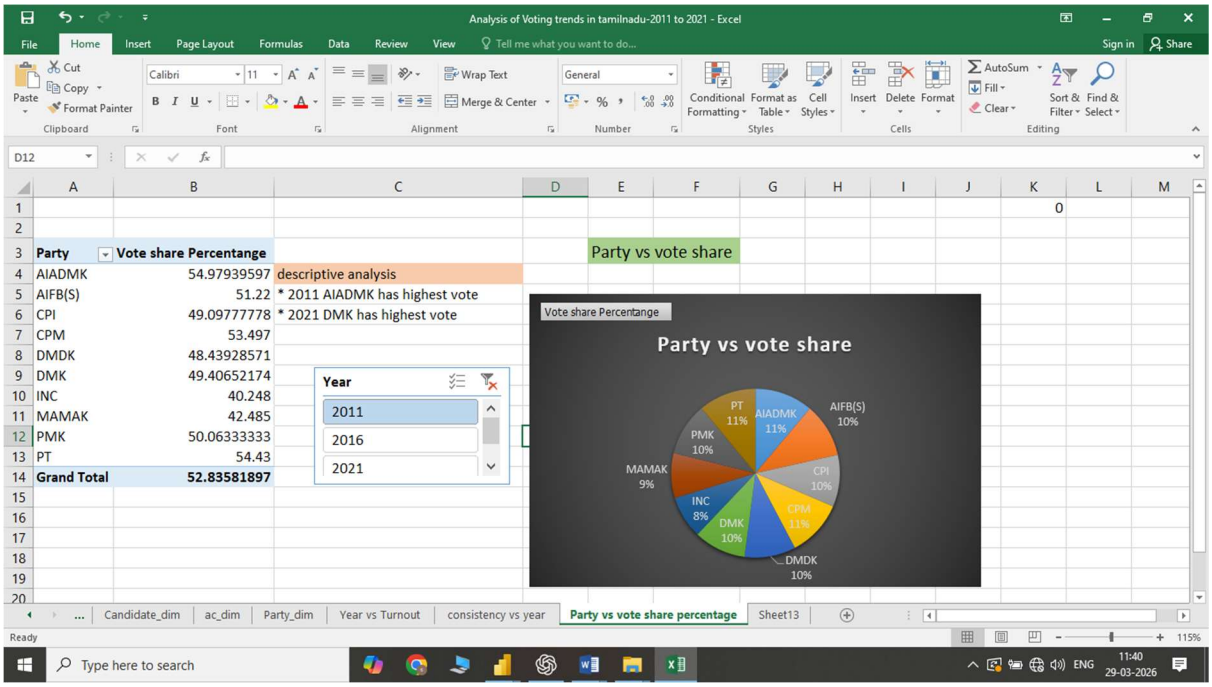
The voter turnout shows a **declining trend** over the three election years:

- **2011:** 78.39% (Highest turnout)
- **2016:** 75.36%
- **2021:** 73.45% (Lowest turnout)

Key Insights:

- There is a **consistent decrease (~5%)** in turnout over a decade.
- Lower turnout may indicate:
  - Reduced political participation among urban voters
  - Decreased interest or trust in electoral processes
- Declining turnout tends to **favor parties with strong loyal voter bases**.

2. Party-wise Vote Share Analysis



The vote share distribution across parties is **closely clustered**, indicating strong competition.

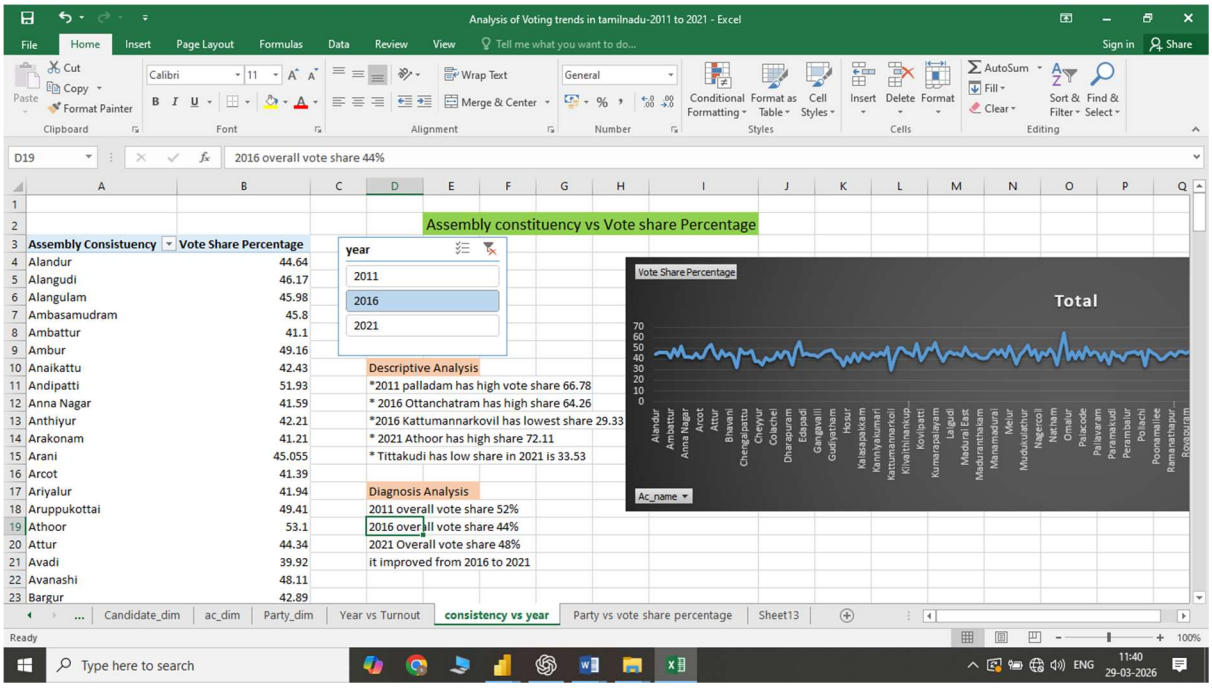
Key Observations:

- Major parties like AIADMK and DMK have vote shares in a similar range (~48%–55%).
- Smaller parties (PMK, CPI, CPM, etc.) contribute around **8–11% each**.

Key Insights:

- The state exhibits a **competitive multi-party system**.
- No single party dominates → elections are **highly competitive**.
- Smaller parties play a **crucial role in alliances**, influencing final outcomes.
- Electoral outcomes are often decided by **coalition strategies** rather than individual party strength.

3. Constituency-wise Vote Share Analysis



Vote share across constituencies shows **high variation**:

- Range: ~29% (lowest) to ~66% (highest)

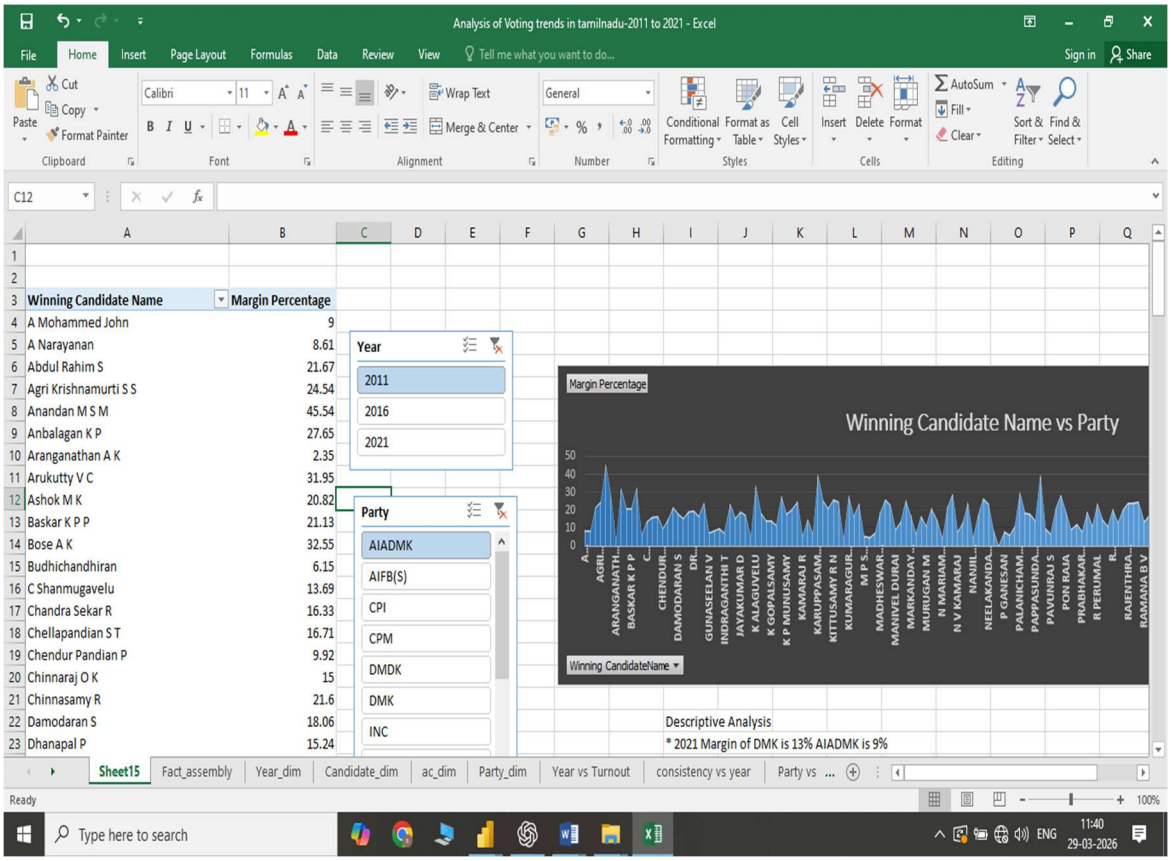
Key Insights:

- Indicates **uneven political influence** across regions.
- Some constituencies act as:
  - **Strongholds** (consistent high vote share)
  - **Swing constituencies** (moderate vote share, highly competitive)
- Regional and local issues significantly impact voting behavior.

4. Winning Margin Analysis

Winning margins vary significantly across candidates:

- Range: Less than 1% to over 35%



Key Insights:

- Many constituencies have **low victory margins (<5%)**, indicating:
  - High competition
  - Sensitivity to small vote changes
- Few constituencies show **high margins**, indicating:
  - Strong candidate influence
  - Party dominance in specific regions

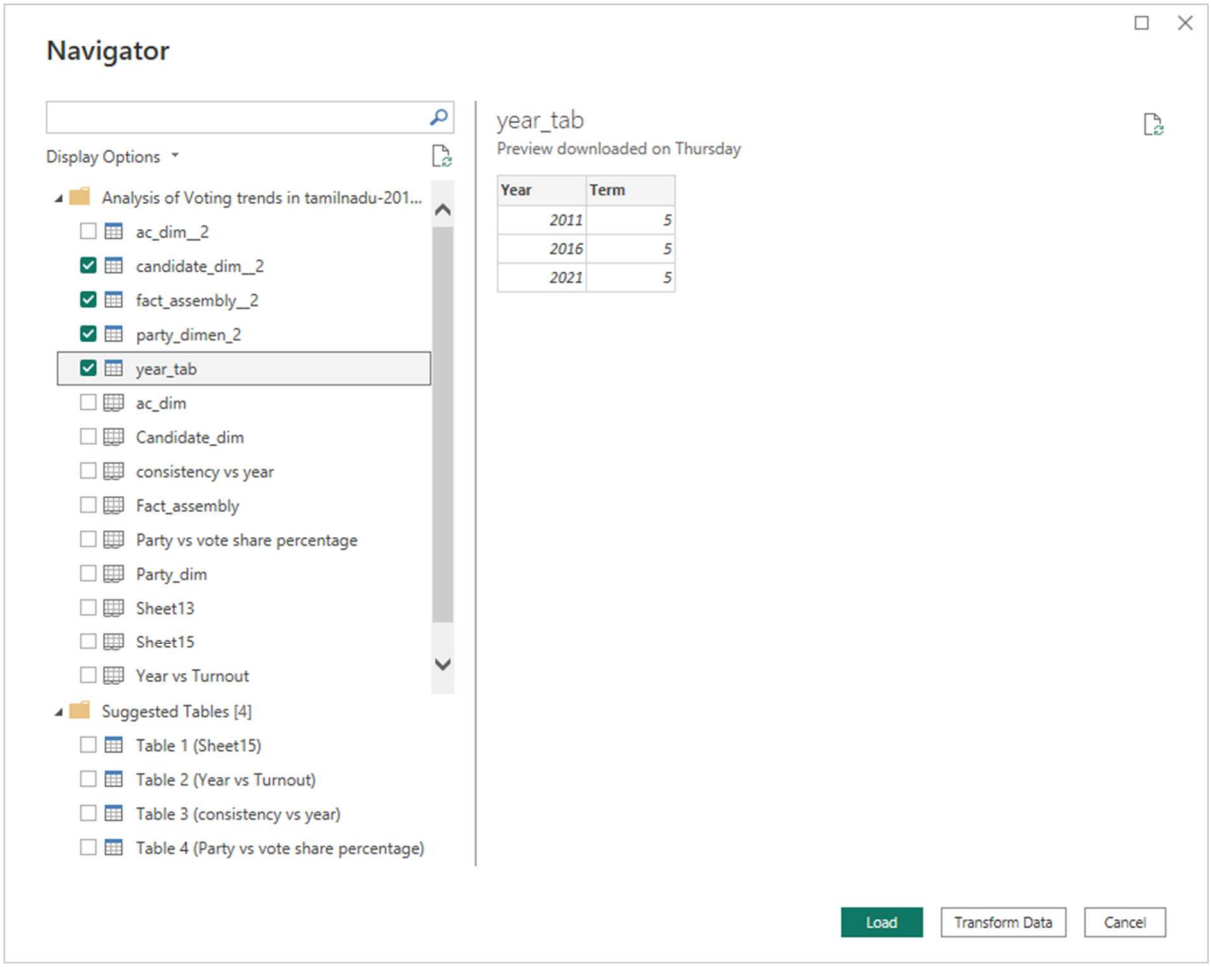
5.Year-wise Trend Analysis

Observations:

- **2016:** Drop in overall vote share (~44%)
- **2021:** Recovery (~48%)

Power BI Project Documentation

1. Data import using Excel Worksheet option



Load and transform data into powerbI,already we cleaned data in Excel .

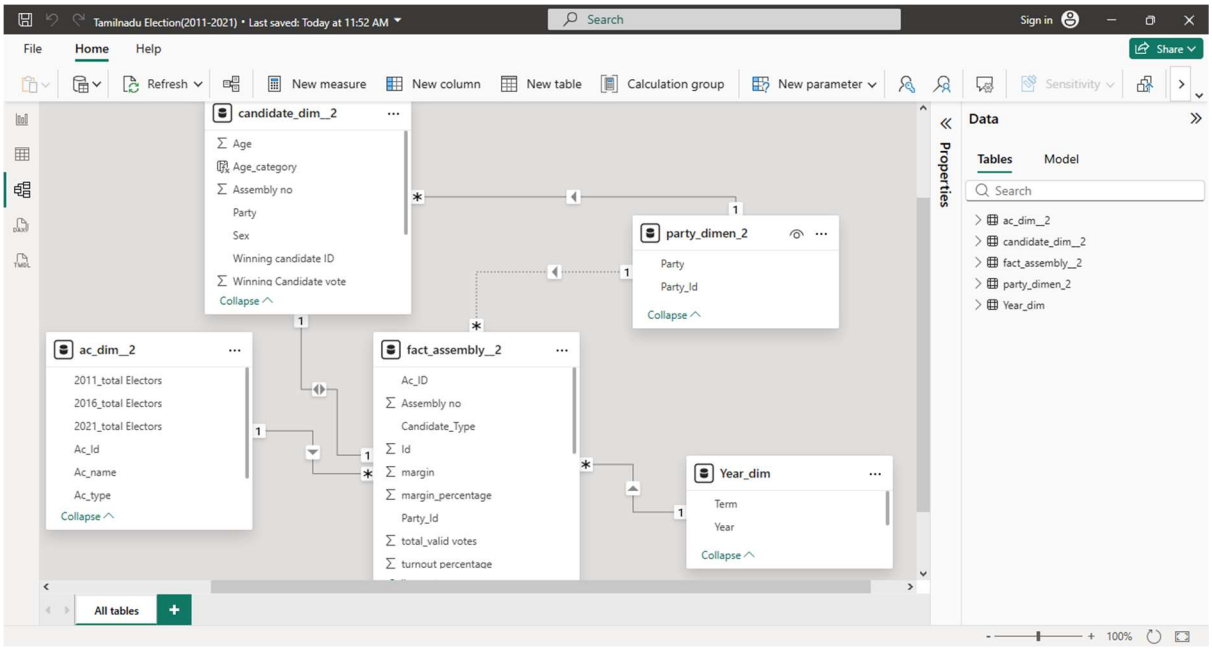
2.Data Sources

The dataset is derived from official election records and structured into multiple tables:

- **ac_dim_2** – Constituency details and total electors per year.
- **candidate_dim_2** – Candidate demographics (age, sex, party, votes).
- **fact_assembly_2** – Fact table containing election results (votes, margins, turnout).
- **party_dimen_2** – Party details with unique IDs.
- **year_dim** – Election year and term mapping.

3.Data Modeling

data model follows a **star schema**:



- **Fact Table:** `fact_assembly_2` (core election results).
- **Dimension Tables:** `candidate_dim_2`, `ac_dim_2`, `party_dimen_2`, `year_dim`.
- Relationships:
  - `fact_assembly_2` ↔ `candidate_dim_2` (Candidate ID).
  - `fact_assembly_2` ↔ `ac_dim_2` (Constituency ID).
  - `fact_assembly_2` ↔ `party_dimen_2` (Party ID).
  - `fact_assembly_2` ↔ `year_dim` (Year).

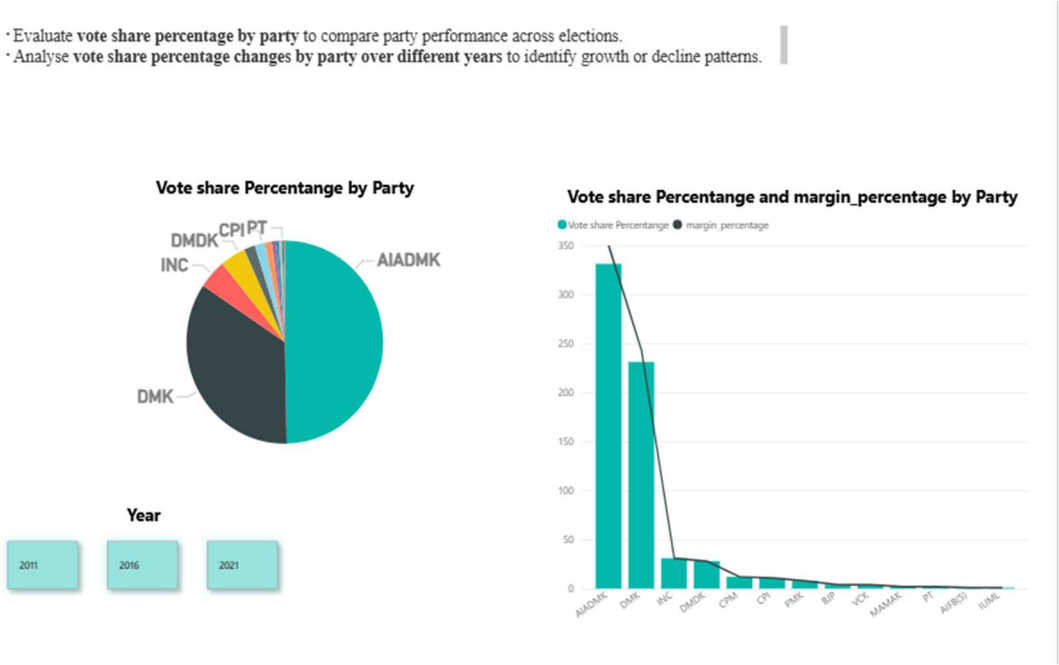
Key Measures (DAX Calculations)

- **Total Votes** = SUM(`fact_assembly_2[total_valid votes]`)
- **Turnout %** = AVERAGE(`fact_assembly_2[turnout percentage]`)
- **Winning Margin** = SUM(`fact_assembly_2[margin]`)
- **Margin %** = AVERAGE(`fact_assembly_2[margin_percentage]`)

Chart-by-Chart Insights:

1.Vote Share Percentage by Party (Pie Chart)

2. Vote Share % and Margin % by Party (Bar/Line Chart)



Key Insights:

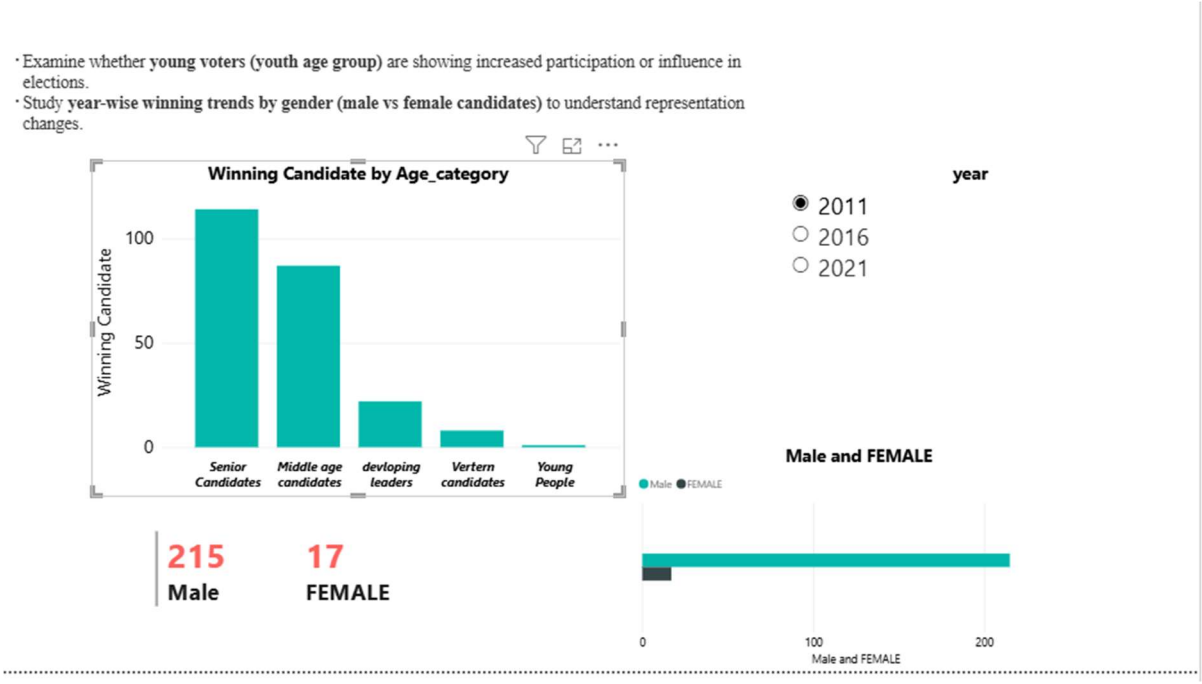
- AIADMK and DMK dominate vote share across years, with smaller parties like INC, CPI, and DMDK contributing marginally.
- The pie chart highlights the **two-party dominance** in Tamil Nadu politics.
- AIADMK shows high vote share with strong margins in earlier years.
- DMK's margin percentage increases significantly in 2021, reflecting a **shift in voter preference**.
- Smaller parties often have vote share but low margins, indicating **limited winning influence**.

3.Winning Candidate by Age Category (Bar Chart)

4. Male vs Female Representation (Bar Chart)

Key insights:

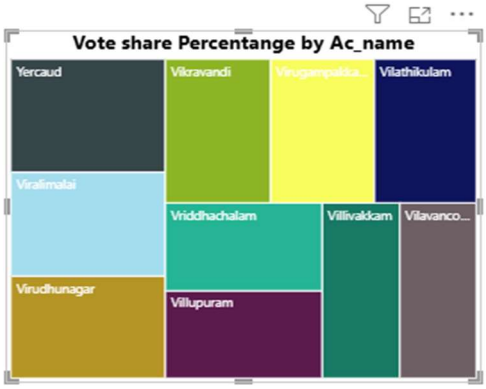
- Senior candidates consistently win the majority of seats.
- Middle-aged candidates form the second-largest group, while **youth representation remains minimal**.



- Male candidates overwhelmingly dominate (200+ wins vs fewer than 25 female wins).
- Gender imbalance persists across all three election years, showing **slow progress in female representation**.

5. Vote Share by Constituency (Treemap)

Identify **which constituencies have the highest vote share** (year-wise and party-wise).



Key insights:

Certain constituencies like Villupuram, Virudhunagar, and Villivakkam show **higher vote shares**, making them politically significant battlegrounds.

- Smaller constituencies contribute less but may still be **strategic swing regions**.

6.Seats by Party (Table)

```
AADIMK_2011_seats = CALCULATE(countrows(fact_assembly__2),fact_assembly__2[year]=2011,fact_assembly__2[Party_Id]="P002")
```

149 AADIMK_2011_seats	23 DMK_2011_SEATS	62 Others
134 AADIMK_2016_seats	88 DMK_2016_SEATS	12 Others1
66 AADIMK_2021_seats	132 DMK_2021_SEATS	36 Other

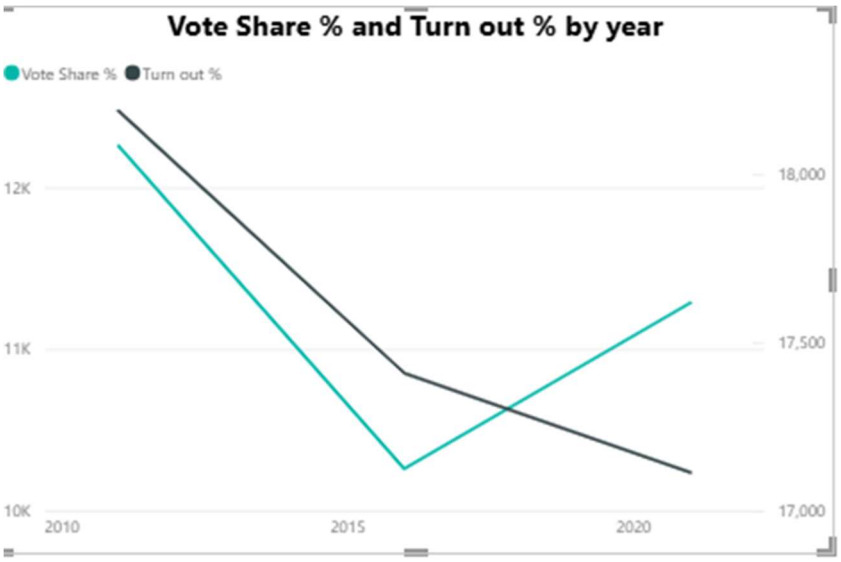
Key Insights:

- AIADMK had strong dominance in 2011 and 2016 but declined sharply in 2021.
- DMK resurged in 2021, securing majority seats.
- "Others" consistently hold a small but **potentially coalition-critical share**.

Turnout %, Vote Share %,

Turnout remains high (70–78%), showing strong voter engagement.

- Both **vote share %** and **turnout %** declined sharply between 2011 and 2016.
- After 2016, the two trends **diverged**:
- **Vote share % increased** (parties consolidated support).
- **Turnout % continued to decline** (fewer voters participated).

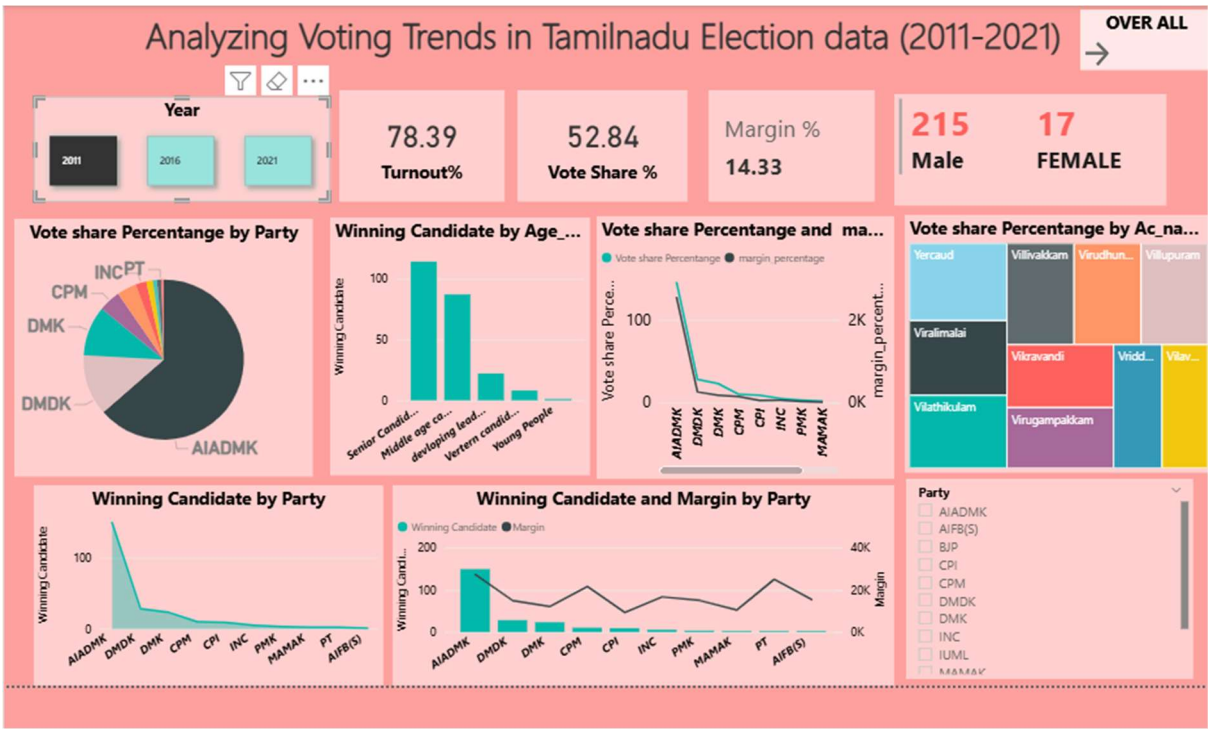


Future Insights

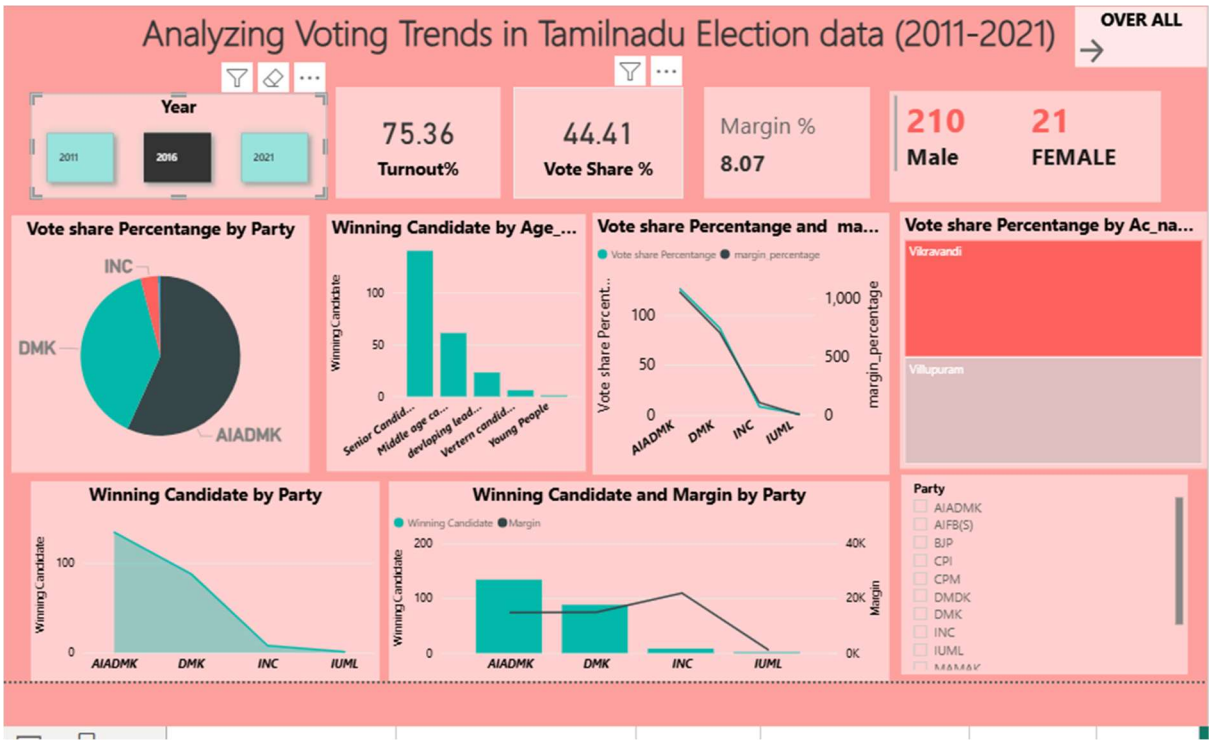
- If turnout continues to decline, elections may become **less representative** of the population.
- **Youth participation** could reverse the turnout decline, especially if parties focus on issues relevant to younger voters.
- **Women's involvement** as candidates and voters could reshape vote share patterns and challenge traditional dominance.
- The chart suggests that **future success depends not only on party strength but also on inclusivity and voter mobilization**.

Final Dash board:

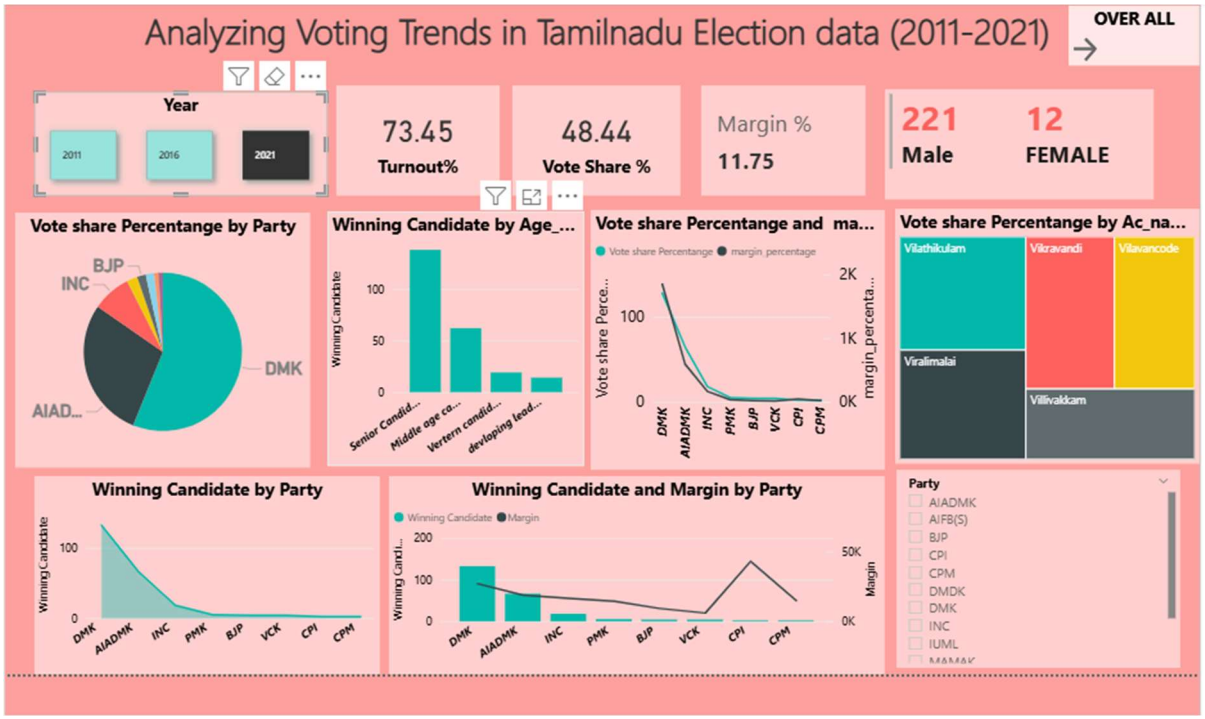
1.Year wise Analysis Dashboard : 2011



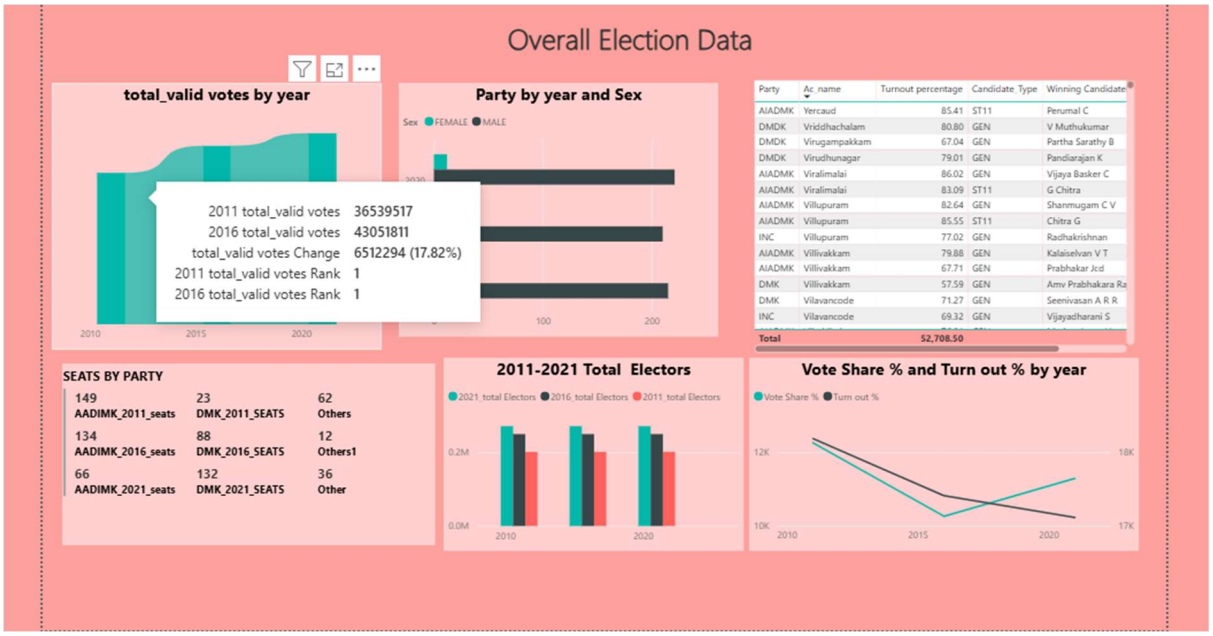
2.Year wise Analysis Dashboard : 2016



3.Year wise Analysis Dashboard : 2021



4.Overall Election Data :



Key Political Insights

- **AIADMK's Dominance (2011 & 2016):**
  The party's strong performance during these years was driven by effective leadership and a well-organized party structure.
- **DMK's Resurgence (2021):**
  The comeback reflects shifting voter sentiment, largely influenced by dissatisfaction with the incumbent government.

Representation Challenges

- **Low Female Representation:**
  This may stem from party nomination practices and persistent cultural barriers that limit women's participation in politics.
- **Struggles of Young Candidates:**
  Voters tend to associate credibility with experience, making it harder for younger candidates to succeed.

Future Trends

- **DMK's Prospects:**
  If current patterns continue, DMK may maintain dominance in upcoming elections. However, margins could narrow as AIADMK regains strength.
- **Women in Politics:**
  Female representation is expected to grow gradually, especially if political parties introduce gender quotas or reforms.
- **Youth Participation:**
  Younger candidates may gain more traction if youth voter turnout increases and parties actively promote generational leadership.

Electoral Dynamics

- **Key Battleground Constituencies:**
  Areas like Villupuram and Virudhunagar, with consistently high vote shares, are likely to remain crucial in future elections.

Broader Democratic Narrative

- **Cycle of Power:**
  The shift from AIADMK dominance to DMK resurgence highlights the cyclical nature of political power—no party remains dominant indefinitely.
- **Voter Trust:**
  Changing electoral outcomes reflect how voters adjust their support based on governance, leadership, and public perception.
- **Participation vs. Inclusion:**
  While voter turnout is high, gaps in representation—especially for women and youth—show that democracy is active but not fully inclusive.
- **Experience vs. New Voices:**
  Senior leaders continue to dominate, indicating voter preference for experience, but this also creates barriers for fresh perspectives.
- **Evolving Democracy:**
  Beyond statistics, electoral trends reveal a deeper story of how democracy evolves, how constituencies become competitive battlegrounds, and how vote margins signal political intensity.

Conclusion:

Tamil Nadu elections from 2011 to 2021 show that power keeps changing between parties. People are very active in voting, which is good for democracy. But women and young people still don’t get enough chances in politics. The future will depend on whether political parties give more opportunities to them.